

sycophancy increases by 1/4 to 1/2 of a point on the Likert scale when the user’s interaction context yields a “very accurate” understanding of their political views compared to when the model has no understanding (abstains from inferring the user’s views).

User demographics are not associated with significant changes in perspective sycophancy. For both models, neither the interaction terms between user demographics and context (β_3 , β_4 , and β_5) nor the demographic variables on their own are statistically significant (Appendix Table 7). This suggests that a model’s understanding of a user’s political views plays a larger role in shaping perspective sycophancy than user demographics themselves. However, demographics may still have an indirect influence by shaping how accurately models infer user views. Our participant pool is not large enough to study interactions between demographics and model understanding, as we do not observe significant differences in understanding by demographics (Appendix Table 8).

5 Discussion

Our analysis shows how interaction context often increases sycophancy in LLMs. In this section, we discuss the limitations of our study as well as its implications for evaluations and system design. Our work indicates that sycophancy is a more complex phenomenon than prior evaluations suggest. There are many forms of sycophancy, each of which may manifest differently depending on the presence and type of interaction context. Moreover, our results suggest that some personalization approaches may amplify sycophancy, raising questions for system design in extended conversations. In particular, we consider: How can systems personalize without amplifying sycophancy? When is sycophancy harmful? What design interventions can reduce sycophancy?

5.1 Limitations

Our study has a few noteworthy limitations. First, our analysis of perspective sycophancy is limited to two models and does not consider synthetic interactions or using memory as context. This is because we measure perspective sycophancy using a post-interaction survey, which had limited questions and was designed to take less than an hour. Second, participants interacted with only one model (GPT 4.1 Mini). While we use the interaction context to evaluate several models, it is uncertain whether our findings would hold if the context itself had been generated by other models. We attempt to mitigate this by excluding queries or responses that mentioned “GPT” or “ChatGPT” from the context used in evaluations. Another limitation is that we cannot directly study the “memory” capabilities of commercial models, since these features are not exposed through the API. In practice, most LLM-based chatbots can “remember” user details across multiple chat sessions, but it remains unclear how these details are extracted from conversation history and provided as context. We evaluate a simple prompt-based method for building memory [51], but commercial methods may be more sophisticated.

Our analysis is further limited by the two-week interaction period with 38 student participants. With a longer interaction period, we hypothesize that models would display stronger mirroring behaviors. Models also show a surprisingly accurate understanding of users’ politics and personality. This limits our ability to compare users where models have “no understanding” of their perspective to

users where models have a “very accurate understanding”, though we control for this in our regression analysis and include zero-shot responses as a baseline. In contrast, interaction lengths and topics vary significantly across participants. While this demonstrates that our results are robust across different users, we cannot causally identify how specific interaction topics influence sycophancy based on our sample size. Finally, we focus on two forms of sycophancy – agreement and perspective-based – but LLMs may mirror users in other ways, such as by adopting their style, tone, or affect. Exploring how interaction context shapes other forms of mirroring is an important direction for future work.

5.2 Implications for Evaluating Sycophancy

Anchoring Evaluations in Context. Our results suggest that previous works may underestimate sycophancy and other model behaviors, given that they conduct evaluations without context. Our evaluation method of prompting models with long conversation histories may not perfectly resemble real-world use either, since users may have multiple chat sessions. However, models should be evaluated with varying context lengths to test robustness and better approximate real-world use. Some contexts may cause models to completely degrade, as we observe with Llama 4 Scout, suggesting that despite supporting a certain context length, some models are brittle and exhibit a “too-many-tokens” effect. Moreover, most LLM-based chatbots distill user conversation history across sessions into a persistent memory profile of the user. Our results show that model behavior can vary greatly depending on whether memory profiles are present. While there is little transparency into how commercial systems build and use memory profiles for personalization, evaluations should consider how personalization can change model responses. Field studies, where evaluation prompts are assessed directly by users in their own interaction windows, may offer the most realistic estimate of model behavior. While our study focuses on the presence and type of context, heterogeneity across users and interaction topics may further introduce variables that shift model behavior. Overall, our work demonstrates why evaluation frameworks must move beyond single-turn or zero-context benchmarks, which fail to capture how model behavior can change in real-world interactions.

Different Forms of Sycophancy. We find that agreement and perspective sycophancy manifest *differently* depending on the interaction context. While agreement sycophancy tends to increase with the presence of any user context, perspective sycophancy requires context that reveals information about the user’s worldview. These results suggest that different forms of sycophancy are distinct phenomena, though further investigation is needed. The literature describes many mirroring behaviors under the umbrella term of “sycophancy,” including flattery, susceptibility to rebuttals, and avoiding disagreement. Each of these behaviors may surface differently across interactions, similar to how sycophancy varies across human relationships. Furthermore, some forms of sycophancy are more explicit, such as agreement, whereas others are more subtle, such as perspective mirroring. As a result, evaluating one form of sycophancy may not reliably provide insights into other forms. Future work should investigate whether different forms of sycophancy stem from shared or distinct underlying mechanisms.

Human Perception of Sycophancy. Evaluations should further investigate how users perceive sycophancy to better understand its consequences. In our analysis of political explanations, we find that roughly half the time, users perceive meaningful semantic differences between responses generated with and without context. Future work should examine whether users also detect such differences in other tasks. While some variation may be expected for political explanations, well-specified or fact-based queries should not produce context-dependent responses that diverge semantically in ways that users can clearly perceive [20]. We also did not investigate the potential downstream effects of context-driven semantic variation – for instance, how perspective sycophancy might shape users’ political beliefs over time. A related direction for agreement sycophancy is to study whether shifts in personal advice influence users’ self-image, mood, or real-world behavior. These questions are increasingly important since information dissemination and personal advice are two of the most common use-cases of LLMs [5, 48].

5.3 Does Personalization Amplify Sycophancy?

Untangling Personalization Approaches. Our results raise the question of whether some personalization approaches may amplify sycophancy. Prior work often attributes sycophancy to preference alignment [29, 35], since users prefer responses that are affirmative or aligned with their perspective. Yet in aligned models, we find that user memory profiles are associated with further increases in agreement sycophancy, and that contexts providing more information about the user drive perspective sycophancy. An important direction for future work is to better understand the mechanisms behind sycophancy and to untangle the role personalization plays. There are many components to personalization, including system prompts, memory profiles, and alignment methods. Each of these components involve varying approaches, such as different instructions in system prompts, different methods for updating memory, and different objectives in alignment. And all of these approaches may further depend on interaction context and what is known about the user. We have a limited understanding of how each of these personalization approaches contribute to sycophancy. For example, one reason why memory profiles did not increase sycophancy in Llama 4 Scout may be that Llama performs poorly at building memory profiles from context. And while user contexts did not increase sycophancy in GPT 5.1, other personalization approaches might.

Non-Sycophantic Personalization. While sycophancy may arise from personalization, models can still personalize their responses in many ways without being sycophantic. In personal advice, models should acknowledge and adapt to a user’s self-image while still offering constructive guidance rather than simply agreeing. Human-based therapy derives much of its value from exposing people to perspectives beyond their own [1], and AI systems that act as “yes-men” risk fostering isolation and undermining therapeutic goals. In political discussions, models should recognize a user’s views but also introduce alternative perspectives when appropriate. If models frame political issues only through the user’s existing worldview, they risk reinforcing the polarization dynamics that already shape information dissemination in news and social media [13, 42]. Our results suggest that mirroring behaviors may be a common way models personalize – and that this often becomes sycophancy in

long-context. Personalization methods should therefore be designed to do more than simple mirroring of user preferences and values.

5.4 Designing Systems To Address Sycophancy

When is Sycophancy Harmful? The first step in addressing sycophancy is identifying when it is harmful for users. We study two forms of sycophancy in specific tasks where the negative implications are evident, but eliminating sycophancy altogether could undermine legitimate forms of personalization and empathetic connection that users value. For example, some degree of perspective mirroring may be beneficial in personal advice but inappropriate in political discussions. Systems should therefore be designed to dynamically address sycophancy across tasks, which requires understanding which forms are harmful and in what contexts. As we discuss above, sycophancy encompasses many mirroring behaviors that may surface differently across interactions. Evaluations of downstream harms should inform how to address these behaviors, but design specifications that articulate how models should handle sycophancy and mirroring across different tasks are also essential.

Design Choices for Context & Memory. Our work suggests that design choices around interaction context and system memory can shape sycophancy. Beyond how context and memory are used for personalization, mechanisms behind how models “remember” details may influence whether users experience sycophancy. In extended conversations, systems should be able to identify only the context or memory details relevant to the task at hand. Yet we observe that some users perceive differences in political explanations generated with and without context, even when models cannot accurately summarize the user’s political views. This suggests that models are either inconsistent in longer contexts or are tailoring responses without any relevant user information. Designing models to better identify relevant details in context and memory may therefore be an important direction for reducing sycophancy.

Interventions to Reduce Sycophancy. Models may be able to automatically detect sycophancy and behavior drift in long-context settings by comparing to zero-shot responses, as our LLM-judge evaluation indicates. Several design strategies could emerge from automated detection of sycophancy. LLMs could actively offer alternative perspectives when they detect excessive agreement, helping users encounter diverse viewpoints and break out of echo chambers. Models could also increase transparency about epistemic uncertainty, explicitly flagging when claims are contested or evidence is limited rather than mirroring users’ perspectives. If a model detects that its responses are becoming overly sycophantic, it could even choose to end the conversation automatically. Instead of detecting and adjusting mirroring behaviors, another approach is to design systems that recognize when user queries allow for many diverse and reasonable responses [20], as in the personal advice and political explanation prompts in our evaluation. For example, in personal advice scenarios asking “Was I in the wrong?”, models could explain both sides (“yes” and “no”). When possible responses span multiple valid perspectives, the pluralistic alignment and social choice literatures [14, 45] may offer strategies for aggregating or selecting among responses, rather than defaulting to the one that most closely matches the user’s viewpoint.

UX Design in Long-Context. In extended interactions, a range of UX design choices could mitigate the impact of sycophancy. Given the stark differences we observe when user memory profiles are present, systems should provide greater transparency around how memory is used. For example, chatbot interfaces could pin the user's memory profile at the top and indicate when and which memory details are used to inform each response. Similarly, when interactions involve long contexts, systems could reference or summarize the specific portions of context used to generate an answer, helping users understand when personalization is occurring. Beyond transparency, interfaces could also allow users to control the degree of personalization itself. In some tasks, users may not want the system to draw heavily from earlier conversation turns or stored memory. Giving users the ability to moderate personalization could reduce sycophancy that emerges from long-context settings.

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